

Negar Haghpahani

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Professional Summary

PhD Student and Graduate Research Assistant in Computer Science specializing in **Edge-AI**, **adaptive inference**, and **hardware-efficient ML**. Designed early-exit frameworks achieving up to **60% energy reduction** with minimal accuracy loss on resource-constrained IoT and wearable devices. Experienced in **deep learning**, **LLM systems**, and **model optimization** across research and production.

Education

Washington State University <i>PhD (Direct Path), Computer Science</i> GPA: 4.0/4.0	Jan 2025 – Present Pullman, WA
Amirkabir University of Technology (Top 3 Nationally) <i>B.Sc. Computer Science</i> CGPA: 3.55/4.0	Sep 2019 – Jul 2023

Technical Skills

ML & Deep Learning: PyTorch, TensorFlow, PyTorch Geometric, Scikit-learn, XGBoost, SHAP, CNNs, LSTMs, Transformers, model compression, quantization

LLM & Generative AI Systems: LangChain, Hugging Face, Transformers, RAG, Agentic AI, On-Device AI Agents

MLOps & Systems: Python, C/C++, SQL, Linux, Git, Slurm/HPC, Apache Airflow, MongoDB

Experience

Washington State University Graduate Research Assistant Pullman, WA	Jan 2025 – Present
- Engineered AdaExit and SEE-RF early-exit architectures, slashing edge device energy consumption by 50–60% with <1% accuracy degradation across 6 wearable-sensor benchmarks - Optimized on-device model inference latency for real-time health-monitoring systems and streamlined deep learning model training cycles on Kamiak HPC clusters via efficient resource allocation	
BizGPT AI Developer Toronto, Canada (Remote)	Apr 2024 – Aug 2024
- Architected an end-to-end LLM-powered chatbot using LangChain and Hugging Face, automating complex enterprise conversational workflows to reduce manual routing overhead - Developed automated ETL pipelines using Apache Airflow, increasing production deployment reliability while minimizing API token consumption and model execution latency	
Bahar Ertebat Gostar Data Engineer	Sep 2022 – Aug 2024
- Built supervised machine learning and clustering pipelines in Python, driving a 15% increase in predictive accuracy for core business intelligence and downstream analytics tasks - Refactored legacy monolithic ML architectures into modular, reusable design patterns, accelerating team development velocity by 35% and optimizing unstructured data streams via MongoDB integration	
Bahar Ertebat Gostar BI Developer Intern	Mar 2022 – Aug 2022
Profiled and optimized 30+ complex SQL queries using SSMS Execution Plans, delivering a 45% reduction in database execution latency and query processing time	

Projects

AdaExit: Adaptive Early-Exit Inference for Edge Devices <i>PyTorch</i>	2025
Designed a confidence-based early-exit framework for adaptive on-device inference, cutting energy 50–60% at <1% accuracy loss across 6 wearable-sensor benchmarks	
EMI: Energy-Aware Early-Exit for Wearable IoT <i>PyTorch, TensorFlow</i>	2026
Applied early-exit inference with Bayesian optimization to derive a Pareto-optimal energy–accuracy frontier and a lookup-table controller for runtime exit selection, deploying AdaExit on NVIDIA GPU and Raspberry Pi edge hardware	
Clinical Digital Twin for Severe TBI Prognosis <i>Python, scikit-learn, PyTorch, XGBoost</i>	2026
Built a digital twin and prognostic pipeline on the TBIMS registry (~17K patients) with missing-data imputation, ensemble feature selection, and 12+ benchmarked ML models, applying reinforcement learning (RL) for sequential decision policies	
LLM-Augmented Next-Action Prediction <i>PyTorch, Transformers</i>	Nov 2025
Benchmarked LSTM vs. LLM-augmented models on CASAS smart-home data, analyzing edge-latency vs. accuracy trade-offs with confusion-matrix and Sankey analysis	

Publications

Haghpahani, N., et al. “SEE-RF: Sensor-Aware Early Exit Random Forests for IoT Devices”	ESWeek 2026
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Teaching Assistant

Washington State University – TA for Advanced Data Structures, Cyber Security, Database Systems	Jan 2025 – May 2026
Amirkabir University of Technology – TA for Numerical Linear Algebra, Principles of Software Design	Sep 2022 – Jul 2024

Awards & Certifications

2nd Place Winner, AgAID Digital AgAthon – Developed predictive ML workflows for crop residue coverage	Feb 2025
NVIDIA Fundamentals of Deep Learning Certification	Apr 2025